

# LUCAS SHANE

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Senior Software Engineer with 9+ years of experience in developing scalable applications across Security, FinTech, Healthcare, and SaaS industries. Expertise in designing microservices architectures, migrating legacy systems, and leveraging technologies like React, Java, Python, AWS, and Kubernetes. Skilled in cloud infrastructure optimization, frontend performance enhancement, and automated testing. Strong in Agile practices, team mentoring, and driving operational efficiency.

## PROFESSIONAL EXPERIENCE

ISI Enterprises, Herndon, VA

03/2023 – 11/2024

### Senior Software Engineer

- Spearheaded development of **Microservices** architecture Security Control Management platform using **React 19**, **Java 17**, **Spring Boot**, **JDBC**, Storybook, WebSocket and AWS services.
- Orchestrated migration of monolithic application (Groovy and Grails) to **React.js** frontend and **Kotlin 1.5** backend Microservices architecture, enhancing maintainability and system flexibility.
- Optimized frontend performance by using React 19/Redux with **Next.js** server-side rendering, boosting page load speed and SEO.
- Engineered scalable APIs using GraphQL, **Spring Boot 2**, and PostgreSQL, integrating CMS functionality with JPA for seamless backend operations and enhancing real-time communication via WebSocket.
- Enhanced Incident Report Feature utilizing React 19, **Python 3.8**, **Django**, React Context API, React Router, **Django REST**, MySQL and Bootstrap.
- Orchestrated **AWS** infrastructure with Lambda, EC2, S3, Cognito, RDS, CloudWatch, CloudFormation and Amplify using Terraform as Infrastructure as Code (IaC).
- Automated testing pipelines, achieving over 95% test coverage using JUnit, Pytest, Cypress, and React Testing Library, reducing productive defects by 30%.
- Collaborated with PM & designers regularly to deliver high-quality products and followed an Agile scrum environment with daily standups to ensure timely and iterative project delivery.
- Mentored junior engineers, establishing best practices and improving team efficiency by 20%.
- Streamlined CI/CD workflows, leveraging Docker, Kubernetes, Maven, and Jenkins, reducing deployment times by 60%.

Jellyfish Technologies, Salt Lake, UT

11/2017 – 12/2022

### Software Engineer 2

- Orchestrated high-performance fintech application using JavaScript, **React**, Material-UI, **Java**, **Spring Boot**, REST APIs, MySQL, Jest, Mocha, JUnit, and Jenkins, improving transaction speeds by 30%.
- Pioneered development of 3 Microservices with Spring Boot 2.0 and **JAX-RS/CXF**, streamlining modularity for key business functions and enhancing performance of CRM queries with Hibernate 5.0.
- Enhanced a people management platform using React, **Node.js**, **Express.js**, D3.js, and Bootstrap while leveraging AWS services (S3, CloudFront, Route 53) to ensure high availability and performance.
- Developed 5 **PowerBI** dashboards that enhanced data visualization, contributing to a 15% improvement in inventory and sales decision-making.
- Designed and executed **ETL** processes with **C#/.NET** for 4 data sources, increasing data accuracy by 25% and processing efficiency by 20%.
- Initiated event-driven architecture with Kafka Streams for CRM analytics, reducing data processing delays by 40% and enabling real-time customer behavior tracking.
- Implemented Redis caching solutions to handle high-traffic spikes, reducing page load times for daily active users.
- Engineered a healthcare SaaS web application using **Vue 2**, **Python 2.7**, **Flask**, **FastAPI**, PostgreSQL, Vuex, Vue Test Utils, Webpack, and Google Cloud Platform, enabling seamless patient management.

- Conducted unit, integration, and end-to-end testing (Cypress, Jest, Mocha, Pytest), ensuring robust functionality with 95%+ test coverage for all assigned projects.
- Followed Agile methodology with daily scrum meetings and providing user story updates with JIRA for user stories and the bug tracking.

Busy Human, Orem, UT

07/2015 – 09/2017

#### Software Engineer | UI Designer

- Refactored a legacy PHP-based CMS for admissions and patient management by migrating to **Java 8** and **Spring Boot 1.5**, enhancing application performance and reliability.
- Pioneered a modular healthcare SaaS application with **Java**, **Spring Boot** and **Angular**, accelerating feature development by 25%.
- Integrated RESTful APIs using **Spring Web** and **Spring Data REST**, increasing request-response efficiency by 15% through optimized API development and caching techniques.
- Redefined architecture of cross-sales platform with **.NET 4.5**, **Angular** components and **Azure** Services and initiated the platform from scratch.
- Developed interactive UI components using **Angular**, Material-UI, and Styled Components, following Material Design guidelines for a modern and responsive user experience.
- Implemented authentication and authorization mechanisms using **OAuth 2.0** and **JWT**, strengthening security across the application.
- Improved frontend performance by optimizing Webpack and Babel configurations, reducing initial page load times by 30% through efficient bundling and code splitting.
- Engaged in pair-programming sessions to learn best practice in software industry and team's coding standard for 6 weeks.

### EDUCATION

Utah Valley University, Orem, UT

07/2011 - 05/2015

#### Bachelor of Computer Science

- Most interesting field: Data Structure and algorithm, Web Development, Software Engineering

### SKILLS & OTHER

Skills: Technical Leadership, Software Design & Architecture, Agile Methodologies, Software Development, Software Engineering Best Practices, Full Software Development Life Cycle, Code Optimization, Roadmap Definition, Object-Oriented Programming, Source Control Management, Test Driven Development, Project Management, Healthcare Industry Experience, Team Building & Management, Status Tracking Frameworks, Quality Assurance, Highly Detail-oriented, Flexible & Adaptable, Self-motivated & Goal-oriented, Dedicated & Committed

Frontend Development: JavaScript, TypeScript, React, Remix, Next.js, Vue.js, Angular, Tailwind CSS, Material UI, Semantic UI, Ant Design

Backend Development: Java, Kotlin, Spring Boot, JDBC, JPA, Hibernate, JEE, J2EE, C#, .NET, Python, Django, Flask, FastAPI, Node.js, Express.js, Nest.js, GraphQL, Apollo Client

Database Management: MySQL, PostgreSQL, MongoDB

Data Visualization: Chart.js, D3.js, Three.js

Content Management Systems: DatoCMS, Builder.io

Testing: JUnit, Pytest, Cypress, React-test-render

State Management: Redux toolkit, Mobx, Redux persist

Cloud Services: AWS, Azure, GCP